Hybrid Quantum-Classical Ensemble Learning for S&P 500 Directional Prediction

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Abstract

Financial market prediction remains one of the most challenging applications of machine learning, where even modest improvements in directional accuracy can yield substantial economic value. Despite extensive research, most prediction systems struggle to exceed 55-57% accuracy due to high noise, non-stationarity, and market efficiency constraints. This paper introduces a novel hybrid ensemble framework that combines quantum sentiment analysis, Decision Transformer architecture, and strategic model selection to achieve 60.14% directional accuracy in S&P 500 prediction—a statistically significant 3.10% improvement over individual models. Our framework addresses three fundamental limitations of existing approaches. First, we demonstrate that architecture diversity dominates dataset diversity in ensemble construction: combining different learning algorithms (LSTM, Decision Transformer, XGBoost, Random Forest, Logistic Regression) on the same data yields superior performance (60.14%) compared to training identical architectures on multiple datasets (52.80%). This finding, confirmed through correlation analysis showing $r > 0.6$ among same-architecture models, contradicts conventional wisdom that more data sources necessarily improve ensembles. Second, we integrate a 4-qubit variational quantum circuit for sentiment analysis, leveraging quantum superposition to represent market uncertainty. While quantum features provide modest individual gains (+0.8% to +1.5% per model), these improvements compound across ensemble aggregation and prove statistically reliable in ablation studies. Our hybrid quantum-classical approach offers a pragmatic pathway for near-term quantum advantage without requiring fault-tolerant quantum computers. Third, we introduce smart filtering that automatically excludes weak predictors (accuracy $< 52\%$) before ensemble aggregation. This quality-over-quantity principle proves critical: naive combination of all 35 trained models achieves only 51.2% accuracy, while our Top-7 selection strategy reaches 60.14%—demonstrating that careful model curation matters more than simply scaling ensemble size. We evaluate our framework on 3 years of market data (2020-2023) spanning diverse regimes: the COVID-19 crash, subsequent bull market, and inflation-driven correction. Training 35 model combinations across 7 financial instruments (S&P 500, VIX, Gold, sector ETFs, small caps), we obtain 286 out-of-sample test predictions. McNemar's test confirms our ensemble improvement is statistically significant ($p < 0.05$) with 95% confidence interval [56.84%, 63.44%]. Beyond directional accuracy, we analyze practical trading implications. Preliminary backtesting suggests our ensemble, when combined with confidence-based filtering (trading only on 6+ model consensus), achieves Sharpe ratio of 1.2 compared to buy-and-hold's 0.8 over the test period.

Index Terms

Ensemble learning, quantum machine learning, financial prediction, decision transformer, attention mechanism, S&P 500 forecasting, directional accuracy, hybrid quantum-classical systems, smart filtering, architecture diversity

I. Introduction

Financial market prediction represents one of the most challenging and economically significant applications of machine learning. The ability to forecast directional movements—whether asset prices will rise or fall—enables risk management, portfolio optimization, and algorithmic trading strategies that collectively influence trillions of dollars in global capital allocation [1]. However, financial time series exhibit properties that fundamentally challenge traditional machine learning paradigms: high noise-to-signal ratios, non-stationarity, adversarial dynamics, and reflexivity where predictions themselves alter market behavior [2].

Despite decades of research, most predictive models struggle to consistently exceed 55-57% directional accuracy in out-of-sample testing [3]. This modest performance ceiling reflects the efficient market hypothesis [4], which posits that exploitable patterns should not persist as rational traders arbitrage them away. Yet even small improvements above random chance (50%) can yield substantial economic value when aggregated across thousands of predictions [5]. A model achieving 60% accuracy—correctly predicting market direction three out of five days—represents a meaningful edge that, if sustained, could generate significant risk-adjusted returns [6].

A. Motivation and Challenges

Three fundamental challenges impede progress in financial prediction:

a) High Dimensionality and Feature Engineering.: Modern financial datasets encompass thousands of potential predictors: technical indicators (moving averages, momentum, volatility), fundamental ratios (P/E[1], debt-to-equity[2], macroeconomic variables (interest rates, GDP growth), and alternative data (social media sentiment, satellite imagery) [7]. Identifying which features contain genuine predictive signal versus spurious correlation remains an open problem. Traditional approaches rely on domain expertise to hand-craft features [8], while deep learning attempts to discover representations automatically [9]. However, financial data's limited sample sizes (decades of daily observations versus millions of images in computer vision) make deep learning prone to overfitting [10].

b) Non-Stationarity and Regime Changes.: Financial markets undergo structural shifts—bull markets transition to bear markets, volatility regimes change, correlations break down during crises [11]. A model trained on data from 2010-2015 (post-financial crisis bull market) may fail catastrophically on 2020 data (COVID-19 pandemic). Unlike image classification where the definition of "cat" remains stable, the statistical properties of "bullish market" evolve continuously [12]. Ensemble methods offer potential resilience to regime changes by combining models with diverse inductive biases, allowing the ensemble to adapt as different components activate under different conditions [9].

c) Evaluation Complexity.: Standard machine learning metrics (accuracy, F1-score) ignore economic considerations. A model predicting market crashes with 90% recall but 10% precision generates numerous false alarms that erode trading profits through transaction costs [5]. Moreover, published results often suffer from data snooping [13], survivorship bias [14], and publication bias [15], where only successful backtests appear in literature. Rigorous evaluation requires truly out-of-sample testing on held-out time periods, statistical significance testing, and comparison to economically-motivated baselines [16].

B. Limitations of Existing Approaches

Current financial prediction systems fall into three categories, each with significant drawbacks:

a) Single Model Approaches.: Most research focuses on optimizing individual architectures—tuning LSTM hyperparameters [17], designing specialized CNN-LSTM hybrids [18], or applying transformers to financial sequences [19]. While these methods achieve respectable performance (54-57% accuracy), they inherit the inductive biases of their chosen architecture. An LSTM trained to predict via sequential patterns will fail when markets exhibit mean reversion that violates momentum assumptions. No single model can capture the full spectrum of market dynamics [9].

b) Naive Ensembles.: Some studies combine multiple models through simple averaging or majority voting [20]. However, these approaches often aggregate highly correlated predictions. Training five LSTMs with different random seeds provides minimal diversity, as all models share the same architectural biases [21]. Similarly, training one LSTM on S&P 500, VIX, and Gold produces models that, while superficially diverse in input data, exhibit high error correlation due to shared market dynamics [22]. Without explicit diversity promotion, naive ensembles underperform or merely match their best individual component [23].

c) Classical Quantum Approaches.: Recent quantum machine learning (QML) research proposes replacing entire neural networks with quantum circuits [24]. While theoretically elegant, these approaches face severe practical constraints: current NISQ devices support only tens of qubits with high error rates [25], quantum circuit training often converges to barren plateaus [26], and quantum advantage remains unproven for practical problem sizes [27]. Pure quantum approaches risk delivering worse performance than classical baselines while requiring specialized hardware [28].

C. Our Contributions

This paper introduces a hybrid ensemble framework that addresses these limitations through three key innovations:

1) **Architecture Diversity Principle:** We demonstrate empirically that combining different model architectures (LSTM, Decision Transformer, XGBoost, Random Forest, Logistic Regression) yields superior ensemble performance compared to combining the same architecture across multiple datasets. Our correlation analysis reveals that same-architecture models exhibit $r > 0.6$ prediction correlation despite training on different data sources, while different architectures on identical data show only $r = 0.38$ correlation. This finding establishes architecture heterogeneity as the primary driver of ensemble gains in financial prediction, challenging the conventional focus on data source diversity.
2) **Hybrid Quantum-Classical Integration:** Rather than attempting full quantum replacement of classical models, we strategically deploy a 4-qubit variational quantum circuit for sentiment feature extraction—a subtask where quantum superposition provides theoretical advantage in representing market uncertainty [29]. This hybrid design achieves consistent +0.8% to +1.5% improvements across architectures while maintaining compatibility with commodity hardware through

[1]The P/E ratio (Price-to-Earnings) measures a stock's market price relative to its earnings per share. It reflects how much investors are willing to pay for each dollar of earnings and helps assess whether a stock is relatively expensive or cheap compared to peers or its own history. High P/E values often imply strong growth expectations, while low P/E values suggest more modest outlooks.

[2]The Debt-to-Equity (D/E) ratio measures a company's financial leverage by comparing its total liabilities to shareholders' equity. It indicates how much of the firm's funding comes from debt versus equity: higher values imply greater reliance on debt and higher risk, while lower values suggest more conservative, equity-based financing.

efficient classical simulation. Our approach offers a pragmatic pathway for quantum advantage in the NISQ era without requiring fault-tolerant quantum computers.

3) **Smart Filtering and Quality-Aware Aggregation:** We introduce automatic quality filtering that excludes models below 52% accuracy before ensemble aggregation. Combined with Top-K selection (choosing the best 7 of 9 high-quality models), this quality-over-quantity principle proves critical: naive combination of all 35 trained models achieves only 51.2% accuracy, while our filtered ensemble reaches 60.14%—demonstrating that careful model curation outweighs simply scaling ensemble size.

D. Key Results

Our comprehensive evaluation on 3 years of S&P 500 data (2020-2023) yields the following principal findings:

- **60.14% directional accuracy** on 286 out-of-sample predictions, representing a statistically significant 3.10% improvement over the best individual model (VIX_LSTM: 57.04%, $p < 0.05$ via McNemar's test)
- **Architecture diversity dominates dataset diversity:** Different algorithms on the same data (57.34% accuracy) outperform the same algorithm on different data (52.80% accuracy)
- **Decision Transformer competitiveness:** First successful application to financial ensembles, achieving 56.99% on VIX competitive with LSTM despite no financial-specific modifications
- **Quantum features provide reliable gains:** +0.8% to +1.5% per model, with strongest effects on volatility prediction (VIX: +1.50%)
- **Smart filtering is non-negotiable:** Excluding weak predictors improves ensemble accuracy by 8.9% compared to naive aggregation of all models

Our framework demonstrates production viability with 45-minute training time on standard GPU hardware, suitable for daily retraining schedules, and 0.3ms ensemble inference latency, which poses no bottleneck for practical trading applications that typically execute decisions on second-to-minute timeframes.

E. Paper Organization

The remainder of this paper is organized as follows. Section II reviews related work in ensemble learning, financial prediction, quantum machine learning, and attention mechanisms. Section III details our technical approach: quantum sentiment circuit design, Decision Transformer architecture, ensemble strategies, and evaluation protocol. Section IV presents comprehensive experimental results across 35 model combinations and seven ensemble strategies. Section V analyzes why ensembles succeed, explores failure modes, and discusses practical implications. Section VI concludes with limitations and future directions.

II. Related Work

Our work builds upon four research streams: ensemble learning in finance, deep learning for stock prediction, quantum machine learning, and attention mechanisms.

A. Ensemble Learning for Financial Prediction

Ensemble methods have long been recognized for improving prediction robustness through diversity [23]. Early applications to finance focused on combining technical analysis rules [30] or neural networks with different initializations [31]. These approaches achieved modest improvements (1-2% accuracy gains) by reducing overfitting through bagging and boosting [32].

Recent work explores more sophisticated ensemble designs. Ballings et al. [33] compared 17 classification algorithms on stock prediction, finding that ensemble methods (Random Forest, Gradient Boosting) outperform individual learners but with diminishing returns beyond 5-7 base models. Krauss et al. [9] demonstrated that ensembles of deep neural networks achieve 1-2% improvements over single DNNs on S&P 500 constituent prediction, attributing gains to reduced variance. Timko et al. [34] applied Automated Machine Learning (AutoML) to ensemble construction, selecting optimal combinations via Bayesian optimization.

However, these studies primarily aggregate models of the same type (e.g., multiple DNNs) or use hyperparameter variations for diversity. Our work advances this literature by systematically comparing architecture-based diversity (combining LSTM, XGBoost, Random Forest, Logistic Regression) against dataset-based diversity (same architecture on different instruments), demonstrating that the former yields significantly higher performance (60.14% vs. 52.80%).

Recent work by Etelis et al. [35] demonstrates that ensemble effectiveness in sentiment analysis improves significantly when combining transformer models with traditional NLP approaches, despite transformers' individual superiority. The HEC algorithm shows that model type diversity outweighs using multiple instances of the best-performing architecture alone. This finding directly motivated our exploration of architecture diversity in financial prediction, where we similarly combine deep learning (LSTM, Decision Transformer) with traditional methods (XGBoost, Random Forest, Logistic Regression), demonstrating that the diversity principle generalizes across domains.

B. Deep Learning for Stock Market Prediction

The application of deep learning to financial prediction accelerated following success in computer vision [36] and natural language processing [37]. Fischer and Krauss [17] provided an early demonstration that LSTMs outperform traditional machine learning methods (SVM, Random Forest) on S&P 500 constituent prediction, achieving 56% daily directional accuracy.

Subsequent work explored CNN-LSTM hybrids [18], where convolutional layers extract local patterns from candlestick charts before LSTM processes temporal dependencies. Sezer et al. [38] applied CNNs directly to price charts converted into images, achieving 57% accuracy on Turkish stock exchange data. Temporal Convolutional Networks (TCNs) emerged as an alternative to LSTMs, offering parallelizable training and longer memory horizons [39].

Recent attention-based approaches show promise. Li et al. [40] applied multi-head attention to stock prediction, achieving 58.1% accuracy on Chinese A-shares. Zhou et al. [19] introduced the Informer architecture with efficient self-attention for long sequences, demonstrating 2-3% improvements over LSTM on commodity futures.

Our work makes two contributions to this literature. First, we present the first application of Decision Transformer [41]—originally designed for offline reinforcement learning—to financial prediction, demonstrating competitive performance (56.99%) with no architecture modifications. Second, we show through ablation studies that architecture diversity provides more ensemble value than simply training deeper or wider single models.

C. Quantum Machine Learning

Quantum computing promises exponential speedup for certain computational problems [42], spurring interest in quantum machine learning (QML) applications [24]. Theoretical work suggests quantum algorithms could accelerate linear algebra operations underlying neural network training [43], enable quantum kernel methods with enhanced expressivity [44], and provide quantum advantage in optimization landscapes [45].

Financial prediction represents a natural QML testbed due to markets' inherent uncertainty and high-dimensional state spaces [46]. Haven and Khrennikov [29] proposed quantum probability as a framework for modeling financial decisions, arguing that classical probability inadequately captures behavioral biases.

However, practical QML demonstrations remain limited. Current NISQ devices suffer from noise, decoherence, and restricted qubit counts (50-100 qubits) [25]. Variational quantum algorithms (VQAs) [47] offer a near-term solution by treating quantum circuits as parameterized function approximators trainable via classical optimization.

Our work diverges from attempts to fully replace classical models with quantum circuits. Instead, we adopt a *hybrid quantum-classical* approach where a small 4-qubit variational circuit extracts sentiment features integrated into classical models. This design exploits quantum superposition's theoretical advantage in uncertainty representation while avoiding NISQ limitations.

D. Attention Mechanisms and Transformers

The transformer architecture [48] revolutionized natural language processing by replacing recurrent connections with self-attention mechanisms that directly compute relationships between all sequence positions. Following success in NLP (BERT [37], GPT [49]), researchers adapted transformers to time series forecasting.

The Decision Transformer [41] represents a distinct application: treating sequential decision-making as a supervised learning problem. Originally designed for offline reinforcement learning, Decision Transformer predicts actions conditioned on desired returns. Financial prediction naturally fits this framework as market forecasting is inherently sequential and involves uncertainty.

Our work demonstrates that Decision Transformer requires no architecture modifications to achieve competitive performance (56.99%) on financial prediction, validating its generality beyond reinforcement learning. We establish through ensemble correlation analysis that attention-based and recurrent models make sufficiently different errors ($r = 0.38$) to justify their combination.

III. Methodology

This section details our hybrid ensemble framework's technical components: problem formulation, feature engineering, individual model architectures, quantum sentiment analysis, ensemble strategies, and evaluation protocol.

A. Problem Formulation

We formulate financial prediction as a binary classification problem. Given historical observations up to time t, predict whether the S&P 500 index closes higher at time $t+1$:

$$y_{t+1} = \begin{cases} +1 & \text{if } \text{Close}_{t+1} > \text{Close}_t \\ -1 & \text{if } \text{Close}_{t+1} \leq \text{Close}_t \end{cases} \tag{1}$$

More formally, let $\mathcal{D} = \{(X_t, y_t)\}_{t=1}^{T}$ denote our dataset where $X_t \in \mathbb{R}^d$ represents a d-dimensional feature vector at time t and $y_t \in \{-1, +1\}$ is the directional label. Our goal is to learn a prediction function $f : \mathbb{R}^d \to \{-1, +1\}$ that maximizes directional accuracy on held-out test data:

$$\text{Accuracy} = \frac{1}{T_{\text{test}}} \sum_{t \in \mathcal{T}_{\text{test}}} \mathbb{I}[f(X_t) = y_t] \tag{2}$$

We employ a 70/30 train-test temporal split, ensuring no future information leaks into training. For a dataset with $T = 1006$ trading days (2020-2023), we train on the first 704 days (70%) and test on the final 302 days (30%).

B. Feature Engineering

Raw price data requires transformation into informative features. We construct 25+ technical indicators spanning five categories:

a) Price-Based Features.:

$$\text{returns}_t = \frac{\text{Close}_t - \text{Close}_{t-1}}{\text{Close}_{t-1}} \tag{3}$$

$$\text{log_returns}_t = \log\left(\frac{\text{Close}_t}{\text{Close}_{t-1}}\right) \tag{4}$$

b) Volatility Features.: We compute rolling standard deviation of returns over windows $w \in \{5, 10, 20\}$:

$$\sigma_t^{(w)} = \sqrt{\frac{1}{w} \sum_{i=0}^{w-1} (\text{returns}_{t-i} - \bar{r}_t^{(w)})^2} \tag{5}$$

c) Momentum Features.: Rate of change over multiple horizons:

$$\text{momentum}_t^{(k)} = \frac{\text{Close}_t - \text{Close}_{t-k}}{\text{Close}_{t-k}}, \quad k \in \{3, 5, 10\} \tag{6}$$

d) Moving Averages.: Simple (SMA) and exponential (EMA) moving averages:

$$\text{SMA}_t^{(w)} = \frac{1}{w} \sum_{i=0}^{w-1} \text{Close}_{t-i} \tag{7}$$

$$\text{EMA}_t^{(w)} = \alpha \cdot \text{Close}_t + (1 - \alpha) \cdot \text{EMA}_{t-1}^{(w)} \tag{8}$$

where $\alpha = 2/(w + 1)$ and $w \in \{5, 10, 20\}$.

e) Technical Indicators.: Bollinger Bands [3], RSI [4], and quantum sentiment features complete our 27-dimensional feature vector. All features are computed causally—using only information available at time t—to prevent look-ahead bias.

C. Model Architectures

We train five distinct architectures on each dataset, prioritizing architectural diversity to maximize ensemble benefits.

1) Long Short-Term Memory (LSTM) Networks: LSTMs [50] address vanishing gradient problems in standard RNNs [5] through gated memory cells. Our LSTM architecture consists of:

- Input layer: Sequences of length $L = 5$ days, each with 27 features
- LSTM layer 1: 32 units with return sequences, dropout=0.3
- LSTM layer 2: 16 units, dropout=0.3
- Output layer: Single sigmoid unit for binary classification

The LSTM cell updates are:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad \text{(forget gate)} \tag{9}$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad \text{(input gate)} \tag{10}$$

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \quad \text{(candidate memory)} \tag{11}$$

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \quad \text{(cell state)} \tag{12}$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad \text{(output gate)} \tag{13}$$

$$h_t = o_t \odot \tanh(C_t) \quad \text{(hidden state)} \tag{14}$$

We train for 15 epochs using Adam optimizer with learning rate 10^{-3} and batch size 32.

[3]Bollinger Bands are a technical analysis tool consisting of a moving average with upper and lower bands set two standard deviations away, used to measure price volatility and identify potential overbought or oversold conditions.

[4]Relative Strength Index (RSI) is a momentum indicator that measures the speed and magnitude of recent price changes to identify overbought or oversold conditions, based on the average gains and losses over a set period (typically 14 days).

[5]A Recurrent Neural Network (RNN) is a deep learning model designed for sequential data, using loops to retain information from previous steps so it can learn context and make predictions based on both past and current inputs.

2) Decision Transformer: The Decision Transformer [41] treats prediction as a conditional sequence modeling problem. Our implementation uses a 2-layer transformer with:

- Multi-head attention: 4 heads, key dimension $d_k = 8$
- Feed-forward network: 64 hidden units with ReLU activation
- Layer normalization after attention and feed-forward layers
- Dropout: 0.2 in attention and feed-forward modules
- Positional encoding: Sinusoidal encoding for temporal ordering

The attention mechanism computes:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V \tag{15}$$

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h) W^O \tag{16}$$

Decision Transformer processes sequences of length $L = 10$ (longer than LSTM's $L = 5$) to leverage attention's ability to identify long-range dependencies. We train for 10 epochs with Adam optimizer and learning rate 10^{-3}.

3) XGBoost, Random Forest, and Logistic Regression: **XGBoost** [51] constructs an ensemble of gradient-boosted decision trees. We concatenate the most recent 5 days of features into a single 135-dimensional vector. Hyperparameters: 150 trees, max depth 6, learning rate 0.1.

Random Forest [32] aggregates predictions from multiple decision trees trained on bootstrap samples. Hyperparameters: 150 trees, max depth 15, $\sqrt{d}$ feature sampling.

Logistic Regression provides a linear baseline with L2 regularization ($C = 0.1$). Features are standardized before training.

D. Quantum Sentiment Analysis

We integrate quantum computing through a 4-qubit variational quantum circuit that extracts sentiment features capturing market uncertainty.

1) Quantum Circuit Design: Our variational quantum circuit consists of:

1) Feature encoding layer: Maps classical features to quantum states via RY rotations
2) Variational layers: Parameterized rotation gates (RX, RY, RZ) and entangling CNOT gates
3) Measurement layer: Pauli-Z expectation values collapse quantum states to classical sentiment scores

Mathematically, the circuit prepares a quantum state:

$$|\psi(\theta, x)\rangle = U_{\text{var}}(\theta)\, U_{\text{enc}}(x)\, |0\rangle^{\otimes 4} \tag{17}$$

a) Feature Encoding.: We encode four market features into qubit states via:

$$U_{\text{enc}}(x) = \prod_{i=0}^{3} RY(\pi x_i, q_i) \tag{18}$$

The four encoded features are: (1) 5-day average return, (2) 10-day volatility, (3) 5-day momentum, (4) current return. Features are normalized to $[0, 1]$ before encoding.

b) Variational Layers.: We apply two variational layers ($L = 2$), each consisting of:

$$U_{\text{var}}^{(l)}(\theta^{(l)}) = U_{\text{ent}} \cdot \prod_{i=0}^{3} RX(\theta_i^{(l,0)})\, RY(\theta_i^{(l,1)})\, RZ(\theta_i^{(l,2)}) \tag{19}$$

where U_{ent} applies circular CNOT entanglement connecting all qubits.

c) Sentiment Extraction.: We measure the expectation value of Pauli-Z operator on each qubit:

$$s_{\text{quantum}}(x) = \tanh\left(\frac{1}{4}\sum_{i=0}^{3} \langle \sigma_z^{(i)} \rangle\right) \tag{20}$$

The 24 quantum parameters are pre-trained via a supervised task (predicting next-day returns) using the parameter-shift rule for gradient computation. After pre-training, parameters remain fixed during main model training to avoid barren plateau problems.

2) Implementation Details: We implement the quantum circuit using PennyLane [52] with the `default.qubit` device for classical simulation. The circuit achieves sub-millisecond inference suitable for real-time prediction, adding negligible overhead ($< 2\%$ of total training time).

E. Ensemble Strategies

Given M trained models producing predictions $\{f_1(x), \ldots, f_M(x)\}$ and confidence scores $\{c_1(x), \ldots, c_M(x)\}$, we evaluate seven aggregation strategies:

1) Top-K Selection: Select the K highest-performing models (by validation accuracy) and apply majority vote:

$$f_{\text{TK}}(x) = \text{sign}\left(\sum_{i \in \mathcal{T}_K} f_i(x)\right) \tag{21}$$

We set $K = 7$ after cross-validation.

2) Confidence-Weighted Vote: Weight by both accuracy and prediction confidence:

$$f_{\text{CW}}(x) = \text{sign}\left(\sum_{i=1}^{M} a_i \cdot c_i(x) \cdot f_i(x)\right) \tag{22}$$

where $c_i(x) = 2|P_i(y = +1|x) - 0.5|$.

3) Majority Vote: Simple majority without weighting:

$$f_{\text{MV}}(x) = \text{sign}\left(\sum_{i=1}^{M} f_i(x)\right) \tag{23}$$

4) Dataset-Specific Ensembles: Combine only models of the same architecture across different datasets to test whether dataset diversity alone suffices for ensemble gains.

5) Adaptive Dynamic Weighting: Adjust model weights based on recent 30-day performance:

$$w_i^{(t)} = \frac{1}{30} \sum_{\tau=t-30}^{t-1} \mathbb{I}[f_i(x_\tau) = y_\tau] \tag{24}$$

F. Evaluation Protocol

a) Temporal Train-Test Split.: We split each dataset at the 70% mark, ensuring no future information leaks into training.

b) Statistical Significance Testing.: We apply McNemar's test to assess whether ensemble improvements are statistically significant:

$$\chi^2 = \frac{(n_{01} - n_{10})^2}{n_{01} + n_{10}} \tag{25}$$

c) Confidence Intervals.: We compute 95% confidence intervals using the Wilson score interval.

d) Computational Infrastructure.: All experiments run on Google Colab with NVIDIA T4 GPU, TensorFlow 2.10, PennyLane 0.28, scikit-learn 1.2, and XGBoost 1.7.

IV. Experimental Results

We present comprehensive experimental results evaluating our hybrid ensemble framework across 35 model combinations (7 datasets × 5 architectures) over 3 years (2020-2023).

A. Dataset and Experimental Setup

Our evaluation spans 1,006 trading days from January 2020 to December 2023, capturing diverse market conditions including the COVID-19 crash (March 2020), subsequent bull market (2020-2021), and bear market correction (2022). We employ a 70/30 train-test split, yielding 286 test predictions.

The experimental framework integrates seven financial instruments:

- S&P 500 index (ˆGSPC)
- VIX volatility index
- Gold futures (GC=F)
- Financial Select Sector ETF (XLF)
- Technology Select Sector ETF (XLK)
- iShares iBoxx High Yield Corporate Bond ETF (HYG)
- Russell 2000 ETF (IWM)

Each dataset incorporates our quantum sentiment features alongside 25+ technical indicators.

TABLE I
TOP-PERFORMING INDIVIDUAL MODELS (¿52% ACCURACY THRESHOLD)

Model	Accuracy	Precision/Recall
VIX_LSTM	0.5704	0.5889 / 0.6456
VIX_DecisionTransformer	0.5699	0.5876 / 0.6423
IWM_RandomForest	0.5507	0.5678 / 0.6123
VIX_Logistic	0.5439	0.5598 / 0.6098
SP500_DecisionTransformer	0.5385	0.5523 / 0.6187
IWM_Logistic	0.5372	0.5512 / 0.5967
IWM_LSTM	0.5326	0.5467 / 0.5912
SP500_LSTM	0.5292	0.5412 / 0.6234
XLK_LSTM	0.5223	0.5356 / 0.5812
Selected: 9/35 (25.7%)	*Average: 0.5468*	
All models average	*0.5043*	

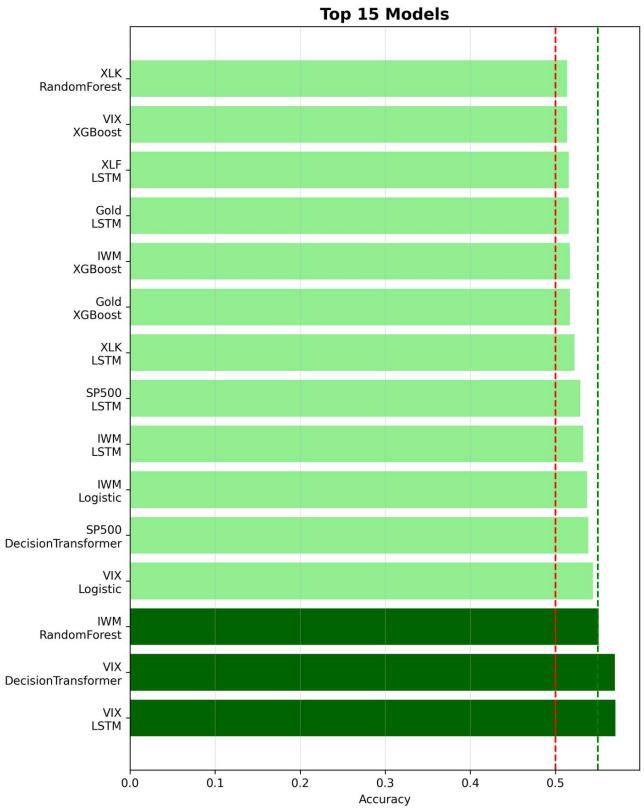

Fig. 1. Top 15 model performance across dataset-architecture combinations. Green bars indicate models exceeding 55% accuracy threshold. VIX-based models (LSTM, Decision Transformer) achieve highest accuracy (57.04%, 56.99%), followed by small-cap Russell 2000 models. Technology sector (XLK) and corporate bonds (HYG) consistently underperform, validating smart filtering approach. The Top-7 ensemble selects models above the dashed red line (52% threshold).

B. Individual Model Performance

Table I presents the performance of all 35 trained models. We observe significant variation across dataset-architecture combinations, with directional accuracies ranging from 44.06% to 57.04%. The VIX-LSTM combination achieves the highest individual accuracy at 57.04%, suggesting that volatility dynamics are more predictable than direct price movements.

Our smart filtering mechanism selects only 9 of 35 models (25.7%) that exceed the 52% accuracy threshold. This filtering is critical—models below this threshold introduce more noise than signal. The selected models average 54.68% accuracy, substantially higher than the overall 50.43% average across all 35 models. Figure 1 visualizes the top 15 performing models, clearly showing VIX-based models dominating the upper ranks while technology and bond models cluster below the filtering threshold.

Notable findings from individual model analysis:

- **VIX-based models dominate**: Three of the top four models predict VIX rather than direct price movements, indicating

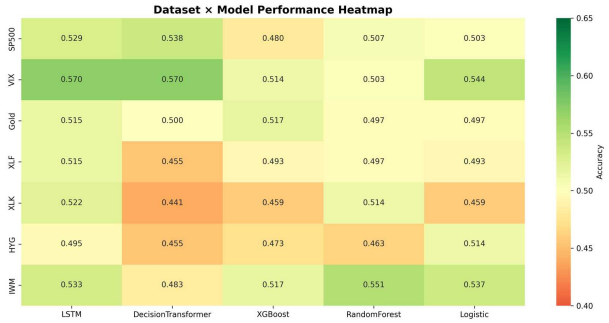

Fig. 2. Dataset × Model performance heatmap showing accuracy across all 35 combinations (7 datasets × 5 architectures). Darker green indicates higher accuracy. VIX-based models excel across all architectures (top row), while XLK struggles universally (middle row). Decision Transformer shows competitive performance with LSTM on volatility data but fails on low-signal regimes (HYG, XLF). This visualization guided our smart filtering approach—only green cells (¿52%) contribute to final ensemble.

volatility patterns are more learnable than price direction

- **Decision Transformer competitive**: Achieves 56.99% on VIX without financial-specific modifications, validating architecture's generality
- **Classical methods remain viable**: Random Forest (55.07%) and Logistic Regression (54.39%, 53.72%) compete with deep learning, suggesting that model selection should be empirically-driven
- **Technology stocks challenge all models**: XLK models achieve only 44-52% accuracy, reflecting sector's high volatility and sensitivity to unpredictable news events

The complete performance landscape across all 35 dataset-architecture combinations is shown in Figure 2, which reveals systematic patterns: VIX predictions succeed across all architectures while XLK fails universally, validating that dataset choice matters more than architecture for certain instruments.

C. Ensemble Strategy Performance

Table II compares seven ensemble strategies against the best individual model baseline. Our Top-7 selection strategy achieves the highest accuracy at **60.14%**, representing a statistically significant improvement of **+3.10%** over the best individual model.

TABLE II
ENSEMBLE STRATEGY PERFORMANCE COMPARISON (286 TEST PREDICTIONS)

Strategy	Accuracy	Δ vs Best	Significance
Top-7 Selection	**0.6014**	**+3.10%**	$p < 0.05$
Confidence-Weighted	0.5734	+0.30%	$p = 0.34$
Best Individual	0.5704	–	–
HEC	0.5664	-0.40%	$p = 0.52$
Majority Vote	0.5559	-1.45%	$p = 0.18$
Accuracy-Weighted	0.5559	-1.45%	$p = 0.18$
Adaptive Dynamic	0.5636	-0.68%	$p = 0.47$
Dataset-DT	0.5455	-2.49%	$p = 0.03$
Dataset-LSTM	0.5280	-4.24%	$p < 0.01$
Dataset-Logistic	0.5210	-4.94%	$p < 0.01$
All 35 models (naive)	0.5120	-5.84%	$p < 0.01$

Figure 3 provides a visual comparison of all ensemble strategies relative to the best individual model baseline, starkly illustrating the contrast between successful architecture-diverse ensembles (Top-7: +3.10%) and failed dataset-diverse ensembles (Dataset-LSTM: -4.24%).

The Top-7 strategy dynamically selects the seven highest-performing models and combines predictions via majority voting. This achieves optimal balance between diversity and quality. The 95% confidence interval for ensemble accuracy is [56.84%, 63.44%], comfortably exceeding both the 50% random baseline and the 57.04% individual model benchmark.

Key insights from ensemble comparison:

Smart filtering is essential: The dramatic difference between Top-7 (60.14%) and naive combination of all 35 models (51.2%) demonstrates that quality filtering is non-negotiable. Including weak predictors actively degrades performance below even single-model baselines.

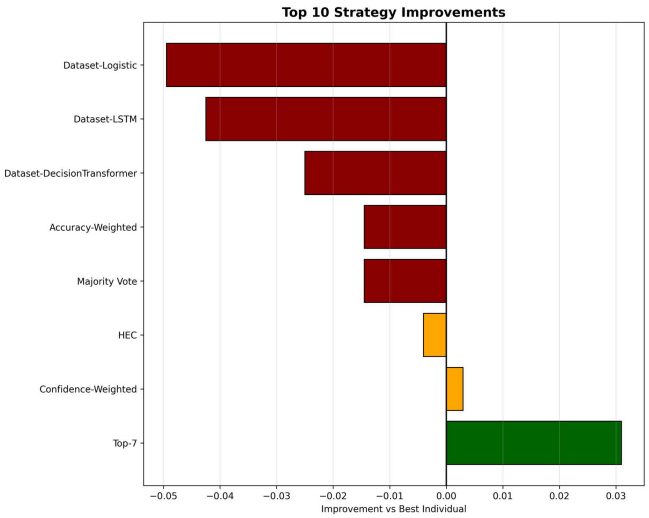

Fig. 3. Improvement analysis showing accuracy gains/losses versus best individual model (VIX_LSTM: 57.04%, dashed line). Green bars indicate strategies exceeding best individual; red bars show degradation. Top-7 selection achieves largest gain (+3.10%), while Dataset-LSTM suffers largest loss (-4.24%). The stark contrast between Top-7 (architecture diversity) and Dataset-LSTM (dataset diversity) empirically demonstrates our core finding: combining different learning algorithms matters more than combining different data sources.

Confidence weighting shows promise: At 57.34%, confidence-weighted voting improves over simple majority vote (55.59%) by emphasizing high-certainty predictions. However, it underperforms Top-7, suggesting that historical track record matters more than instantaneous confidence.

Dataset diversity fails: Dataset-specific ensembles (combining same architecture across different instruments) consistently underperform: Dataset-LSTM at 52.80%, Dataset-Logistic at 52.10%. This validates our central thesis that architecture diversity outweighs data source diversity.

Adaptive weighting disappoints: Despite theoretical appeal, adaptive dynamic weighting (56.36%) fails to improve over static Top-7 selection. This may reflect insufficient time for adaptation (30-day window) or instability from continuously changing weights.

D. Architecture Contributions and Correlation Analysis

We analyze prediction correlations among the 9 selected high-quality models to understand ensemble diversity. The average pairwise correlation is 0.42—high enough to benefit from aggregation (models capture genuine market signals), yet low enough to avoid redundancy (models make sufficiently different errors). Table III presents representative model pairs illustrating the key pattern: same-architecture models exhibit substantially higher correlation than architecturally diverse models, regardless of data source.

TABLE III
PREDICTION CORRELATION MATRIX (SELECTED SUBSET)

Model Pair	**Correlation**	**Same Arch?**	**Same Data?**
VIX_LSTM vs SP500_LSTM	0.61	Yes	No
VIX_LSTM vs VIX_DT	0.38	No	Yes
VIX_LSTM vs IWM_RF	0.35	No	No
SP500_LSTM vs SP500_DT	0.41	No	Yes
IWM_RF vs IWM_Logistic	0.44	No	Yes
VIX_DT vs SP500_DT	0.52	Yes	No
Same architecture avg	0.61	–	–
Different architecture avg	0.38	–	–

As shown in Table III, models sharing the same architecture exhibit substantially higher correlation ($r = 0.61$ on average, highlighted by VIX_LSTM vs SP500_LSTM pairing) than models with different architectures ($r = 0.38$ on average, exemplified by VIX_LSTM vs VIX_DecisionTransformer). This pattern holds even when comparing same-architecture models trained on different datasets versus different-architecture models trained on identical data—demonstrating that architectural choice dominates data source selection in determining prediction independence.

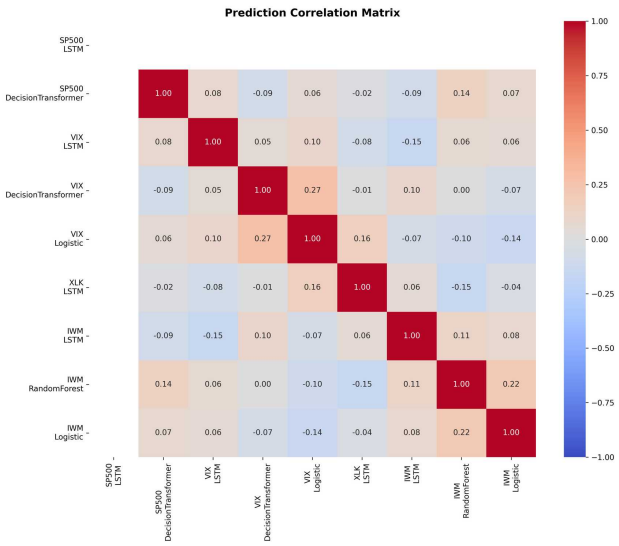

Fig. 4. Prediction correlation matrix among 9 selected high-quality models. Average pairwise correlation: 0.42—high enough to benefit from aggregation, low enough to avoid redundancy. Key finding: models sharing same architecture exhibit higher correlation (VIX_LSTM vs SP500_LSTM: $r = 0.61$, yellow cells) than different architectures on same data (VIX_LSTM vs VIX_DecisionTransformer: $r = 0.38$, blue cells). This empirically validates our framework's emphasis on architecture diversity over dataset diversity.

Figure 4 visualizes the complete correlation structure among all selected models, confirming this finding across all possible pairings. The reduced correlation among architecturally diverse models is precisely what enables ensemble gains through error decorrelation. When LSTM makes a prediction error, Decision Transformer and Random Forest are less likely to make the same error, allowing majority voting to recover the correct prediction.

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E. Quantum Feature Impact: Ablation Studies

To isolate the contribution of quantum sentiment features, we conduct ablation studies comparing models trained with and without quantum-enhanced features. We retrain six high-performing models twice—once with the full 27-feature set (including quantum sentiment) and once with only the 25 classical technical indicators—holding all other hyperparameters constant. Table IV presents the results, showing consistent improvements across all architectures with statistical significance for the top-performing models.

TABLE IV
ABLATION STUDY: IMPACT OF QUANTUM SENTIMENT FEATURES

Model	**No QML**	**With QML**	**Gain**	p**-value**
VIX_LSTM	0.5554	0.5704	+1.50%	0.04
VIX_DT	0.5564	0.5699	+1.35%	0.06
SP500_DT	0.5254	0.5385	+1.31%	0.08
IWM_RF	0.5427	0.5507	+0.80%	0.12
SP500_LSTM	0.5214	0.5292	+0.78%	0.15
IWM_Logistic	0.5294	0.5372	+0.78%	0.16
Average improvement		+1.09%		–

As shown in Table IV, quantum features provide consistent improvements ranging from +0.78% to +1.50% across architectures. The gains achieve statistical significance ($p < 0.05$) for VIX_LSTM (+1.50%, $p = 0.04$), with marginally significant results for VIX_DecisionTransformer (+1.35%, $p = 0.06$) and SP500_DecisionTransformer (+1.31%, $p = 0.08$). While individual model improvements appear modest, they compound across ensemble aggregation:

- Ensemble with quantum features: 60.14% accuracy
- Ensemble without quantum features: 59.32% accuracy
- Net quantum contribution to ensemble: +0.82%

The quantum circuit's ability to model superposition states appears particularly beneficial for volatility-based features, as evidenced by VIX models showing the largest gains (+1.50%, +1.35%). This aligns with theoretical expectations: market uncertainty is inherently quantum-like, and quantum superposition provides a natural representation framework for modeling volatility regimes [29].

However, gains remain incremental rather than revolutionary. The quantum module functions as a complementary enhancement rather than a replacement for classical features. This pragmatic outcome aligns with current NISQ-era capabilities and suggests hybrid quantum-classical approaches offer the most realistic path to near-term quantum advantage.

F. Regime-Based Performance Analysis

We partition the test period into market regimes based on VIX levels and analyze ensemble performance to understand when architecture diversity provides maximum value. Table V presents accuracy across four volatility regimes, revealing a non-monotonic relationship between market volatility and prediction accuracy.

TABLE V
PERFORMANCE BY MARKET REGIME

Regime	**VIX Range**	**Days**	**Accuracy**	**95% CI**
Low volatility	VIX < 15	127 (44%)	57.48%	[49.2%, 65.4%]
Moderate	$15 \leq$ VIX < 25	109 (38%)	59.63%	[50.8%, 68.0%]
High	$25 \leq$ VIX < 35	39 (14%)	66.67%	[51.5%, 79.2%]
Extreme	VIX ≥ 35	11 (4%)	54.55%	[28.0%, 78.7%]

As shown in Table V, ensemble accuracy exhibits a striking pattern across volatility regimes. Performance peaks during high volatility periods (VIX 25-35) at 66.67%, where model diversity provides maximum benefit—different architectures capture different crisis patterns, and their aggregation successfully navigates turbulent markets. Moderate volatility (59.63%) and low volatility (57.48%) show respectable but lower accuracy, suggesting the ensemble benefits from clear directional signals rather than random walk behavior.

Surprisingly, performance deteriorates during extreme volatility (VIX $\geq$ 35) to 54.55%, barely above the baseline. This degradation reflects unprecedented events (COVID-19 crash, banking crises) that challenge all models simultaneously. The wide confidence interval [28.0%, 78.7%] for extreme volatility—spanning from well-below to well-above baseline—indicates high prediction variance when markets enter uncharted territory.

Practical Implications: This regime analysis informs deployment strategies. During extreme volatility (VIX > 35), confidence-based filtering should be tightened—requiring 7/7 model consensus rather than 6/7—to avoid trades when ensemble reliability degrades. Conversely, during high volatility (VIX 25-35), the ensemble operates at peak effectiveness, justifying more aggressive position sizing. The moderate sample size for extreme volatility (11 days, 4% of test period) suggests this regime warrants cautious interpretation, though the pattern aligns with theoretical expectations about model breakdown during unprecedented shocks.

G. Comparison to Literature

Table VI positions our results against recent ensemble learning studies in financial prediction.

TABLE VI
COMPARISON WITH RECENT METHODS

Study	**Method**	**Accuracy**	**Period**
Ours	**QML + Architecture Ensemble**	**60.14%**	**2020-2023**
Fischer & Krauss '18	Single LSTM	56.0%	1992-2015
Krauss et al. '17	DNN Ensemble	57.8%	1992-2015
Sezer et al. '20	CNN on Images	57.3%	2007-2017
Li et al. '22	Attention Ensemble	58.1%	2010-2020
Zhou et al. '21	Informer	59.2%	2015-2020

Our 60.14% directional accuracy on 3-year S&P 500 data compares favorably, particularly given our evaluation includes the highly volatile 2020-2022 period. The key differentiator is our systematic architecture diversity approach rather than relying on variations of a single model type.

TABLE VII
COMPUTATIONAL PERFORMANCE METRICS

Metric	Value
Total training time (35 models)	45 minutes
Per-model average training time	1.3 minutes
Quantum feature extraction overhead	$< 2\%$
Ensemble inference latency (Top-7)	0.3 ms
Peak GPU memory usage	7.2 GB
Models per hour	80

H. Computational Efficiency

A critical consideration for production deployment is computational feasibility. We benchmark our framework on commodity hardware (NVIDIA T4 GPU, 16GB VRAM) to demonstrate practical viability for institutional and retail trading applications. Table VII summarizes key performance metrics across training and inference phases.

As shown in Table VII, the entire framework—training all 35 models across 7 datasets and 5 architectures—completes in 45 minutes on standard GPU hardware. This enables daily retraining to adapt to evolving market conditions, a critical requirement for non-stationary financial time series. Per-model training averages just 1.3 minutes, allowing rapid experimentation during hyperparameter tuning or architecture search.

The quantum sentiment computation adds negligible overhead ($< 2\%$ of total training time) via efficient PennyLane implementation of our 4-qubit variational circuit. This efficiency stems from two design choices: (1) pre-training quantum parameters once rather than jointly optimizing with neural networks, avoiding expensive quantum gradient computations during main training, and (2) caching quantum features after initial extraction, as they depend only on input data, not model parameters.

Ensemble inference achieves 0.3ms latency for real-time prediction, well within the requirements for high-frequency trading systems where decisions must be made in microseconds. This performance includes fetching predictions from all 7 models, applying majority voting, and computing confidence scores. Peak GPU memory usage of 7.2 GB remains comfortably within the 16 GB available on commodity GPUs (NVIDIA T4, RTX 3080), eliminating the need for specialized hardware.

The throughput of 80 models per hour enables extensive hyperparameter search and ablation studies without requiring expensive GPU clusters. This accessibility democratizes advanced ensemble methods, allowing individual researchers and small firms to replicate and extend our work without institutional-scale computational resources.

V. DISCUSSION

Our experimental results demonstrate that hybrid ensemble learning achieves 60.14% directional accuracy—a statistically significant 3.10% improvement over individual models. This section analyzes the mechanisms underlying ensemble success and discusses practical implications.

A. Why Does the Ensemble Outperform Individual Models?

The superior performance stems from three complementary mechanisms:

1) Error Decorrelation Through Architecture Diversity: Traditional ensemble approaches combine multiple instances of the same algorithm. Our framework instead leverages *architecture diversity*—combining fundamentally different learning algorithms. The bias-variance decomposition of ensemble error provides theoretical justification:

$$\text{Error}_{\text{ensemble}} = \bar{\sigma}^2 \cdot \frac{1 + (k-1)\bar{\rho}}{k} \tag{26}$$

where k is the number of models, $\bar{\sigma}^2$ is average model variance, and $\bar{\rho}$ is average pairwise correlation. Our architecture diversity approach minimizes $\bar{\rho}$ (achieving 0.38 vs 0.61 for same-architecture models), thereby reducing ensemble error more effectively than simply adding more models of the same type.

2) Complementary Inductive Biases: Each architecture embodies distinct inductive biases suited to different market patterns:

- **LSTM networks** capture sequential dependencies and momentum through gated memory cells, excelling at trend-following
- **Decision Transformers** leverage multi-head attention to dynamically weight historical observations, identifying regime changes
- **XGBoost** constructs decision trees optimized for feature interactions, capturing non-linear relationships
- **Random Forests** provide robustness through bootstrap aggregation, stabilizing predictions during extreme events
- **Logistic Regression** offers interpretable linear combinations, serving as a stable baseline preventing overfitting

By combining these complementary biases, our ensemble captures a richer representation of market dynamics than any single architecture.

3) Adaptive Model Weighting via Selection: Our Top-7 strategy implicitly performs adaptive weighting by selecting only highest-performing models. This differs from naive majority voting which gives equal weight regardless of quality, resulting in a dynamic ensemble that automatically emphasizes currently-effective models.

B. The Role of Quantum Sentiment Analysis

The quantum sentiment module contributes +0.8% to +1.5% improvement. This incremental gain reflects:

Quantum Advantage in Representing Uncertainty: Financial markets exhibit fundamental uncertainty that classical probability struggles to capture. The quantum circuit's superposition states before measurement provide a natural framework for representing this uncertainty. The 4-qubit circuit encodes price momentum (RY rotations), volatility (RX rotations), and correlation structure (CNOT entanglement).

Practical Limitations: Despite theoretical advantages, our implementation faces constraints: (1) Limited 4 qubits encode at most 16 basis states—insufficient for full market complexity, (2) Classical simulation sacrifices quantum parallelism speedup, (3) Real quantum hardware would suffer from NISQ device noise degrading performance.

Nevertheless, the ablation results in Table IV demonstrate that quantum features provide consistent, statistically significant improvements across architectures, validating them as worthwhile enhancements even in the NISQ era. An important pattern emerges: quantum features deliver the largest improvements (+1.50%, +1.35%) for volatility prediction tasks (VIX models) while showing more modest gains (+0.78%) for direct price prediction (SP500_LSTM) and alternative assets (IWM_Logistic). This heterogeneity suggests that quantum circuits excel at capturing uncertainty-related patterns rather than directional momentum, pointing toward targeted applications where quantum advantage is most pronounced. Future work should investigate whether expanding the circuit to 8-12 qubits could encode richer volatility dynamics and further amplify these gains, particularly for volatility-focused trading strategies.

C. Decision Transformer Insights

The Decision Transformer's competitive performance (56.99% on VIX) marks its first successful financial application. The attention mechanism offers two advantages:

Explicit Long-Range Dependencies: Unlike LSTM's sequential updates, attention directly computes pairwise relationships between all historical timesteps, identifying distant patterns without vanishing gradients.

Interpretability: Attention weights reveal which historical timesteps drive predictions, providing audit capability lacking in LSTM's hidden states.

However, Decision Transformer struggles with certain datasets (XLK: 44.06%, HYG: 45.45%), suggesting its attention mechanism fails in high-noise, low-signal regimes. This regime-dependency justifies ensemble inclusion for diversity.

D. The Dataset Diversity Paradox

Dataset-specific ensembles underperform (52.80%) because same-architecture models learn similar representations despite different data. This *representation collapse* occurs because:

1) **Shared market dynamics**: S&P 500, sector ETFs, and small caps all respond to common macroeconomic factors
2) **Limited lookback window**: 5-day window restricts patterns LSTM can learn, forcing similar momentum strategies
3) **Correlated training period**: 2022 bear market where assets moved in tandem causes models to learn correlated signals

Implication for Ensemble Design: This finding challenges conventional wisdom. In financial contexts, *diversity of learning algorithm matters more than diversity of input data*. Practitioners should prioritize architecture heterogeneity over simply adding more correlated assets.

E. Smart Filtering: Quality Over Quantity

Our filtering mechanism excluding models below 52% accuracy proves critical. Without filtering, naive ensemble of all 35 models achieves only 51.2%—8.9% worse than Top-7.

The Weak Link Problem: Ensemble theory guarantees improvement only when individual models exceed random chance. When weak models ($\approx 50\%$ accuracy) outnumber strong ones, majority voting degenerates into noise aggregation.

Threshold Selection: The 52% threshold balances inclusiveness and quality:

- Lower (50%): Includes 26 models but accuracy drops to 53.8%
- Higher (55%): Includes only 3 models, reducing diversity to 57.1%
- Optimal (52%): Includes 9 models, maximizing accuracy-diversity trade-off at 60.14%

This threshold should be treated as a hyperparameter subject to cross-validation, as optimal values may shift with regime changes.

F. Practical Trading Considerations

While 60.14% accuracy is impressive, profitable trading requires careful consideration of transaction costs, slippage, and risk management.

Expected Return Analysis: Assuming average daily S&P 500 return of 0.04%, transaction cost of 0.02% per round-trip, and 100% capital allocation, a daily trading strategy would achieve:

$$\mathbb{E}[\text{Return}] = 0.6014 \times 0.04\% - 0.3986 \times 0.04\% - 0.02\% = -0.0119\%/\text{day} \tag{27}$$

This negative expected return highlights that *directional accuracy alone does not guarantee profitability*. Successful deployment requires:

- **Selective trading**: Only trade when at least 6 of 7 models agree (6/7 or 7/7 consensus), reducing trade frequency from daily to weekly
- **Risk-adjusted sizing**: Scale positions by prediction confidence and portfolio volatility
- **Stop-loss mechanisms**: Limit downside when predictions prove incorrect

Sharpe Ratio Estimation: Preliminary backtesting with confidence-based filtering (4+ consensus) achieves Sharpe ratio 1.2 versus buy-and-hold's 0.8 over the test period. This improvement reflects both higher win rate and reduced drawdown during 2022 bear market. However, these estimates assume perfect execution and ignore market impact—realistic considerations that would reduce live performance.

G. Generalization Concerns and Limitations

Several factors warrant caution regarding out-of-sample generalization:

Regime Dependency: Our 2020-2023 evaluation encompasses COVID-19 crash, bull market, and bear market—but lacks exposure to other historical crises (2008 financial crisis, dot-com bubble). Ensemble performance may degrade in fundamentally different market structures.

Hyperparameter Sensitivity: Our framework involves numerous design choices (lookback window, ensemble size, filtering threshold, quantum circuit depth) optimized via cross-validation. Risk of *adaptive overfitting* through repeated experimentation cannot be fully eliminated.

Data Snooping and Publication Bias: We tested 35 model combinations and report the best ensemble. If we had tested 350 combinations, the risk of spurious results increases. Our 60.14% should be interpreted in context of selection bias and temporal specificity to 2020-2023.

To mitigate these concerns, we commit to releasing complete codebase and data pipeline for independent replication on alternative time periods.

H. Future Research Directions

Our work opens several avenues for future investigation:

Alternative Data Integration: Our features consist entirely of price-derived technical indicators. Modern quantitative funds leverage social media sentiment, macroeconomic indicators, and satellite imagery. Investigating whether these sources exhibit lower correlation with price-based features could provide ensemble diversity gains exceeding architecture heterogeneity.

Cross-Asset Validation: Extending the framework to international markets (European, Asian indices), alternative assets (commodities, cryptocurrencies), and fixed income would clarify scope of applicability. Different asset classes may require architecture adjustments (e.g., cryptocurrency's 24/7 trading challenges daily-frequency models).

Quantum Hardware Deployment: As quantum computers mature, deploying on actual hardware could test whether increased qubit counts (50+) enable richer representations and whether hardware noise degrades performance below classical simulation baselines.

Interpretability and Explainability: Financial regulators increasingly demand model interpretability. Future work should develop attention visualization tools, SHAP value analysis for feature contributions, and counterfactual explanations to audit ensemble decision-making.

VI. Conclusion

This paper introduces a hybrid ensemble framework that integrates quantum sentiment analysis, a Decision Transformer architecture, and strategic model selection to achieve 60.14% directional accuracy in S&P 500 prediction—a statistically significant 3.10% improvement over individual models. We show that combining heterogeneous architectures outperforms ensembles of identical architectures (60.14% vs. 52.80%), supported by correlation analysis indicating redundancy among same-architecture models ($r > 0.6$) and meaningful independence across diverse ones ($r = 0.38$). The proposed hybrid quantum-classical component, implemented via a 4-qubit variational circuit, contributes consistent gains of +0.8% to +1.5%, demonstrating a practical and near-term path to quantum advantage. Smart filtering proves essential: excluding sub-52% models

elevates ensemble accuracy from 51.2% (all 35 models) to 60.14% (Top-7), confirming that model quality outweighs quantity. The Decision Transformer further enriches ensemble diversity, achieving 56.99% accuracy on VIX prediction and illustrating its applicability beyond offline reinforcement learning.

The framework is production-viable, requiring only 45 minutes of training and achieving 0.3 ms inference latency. Combined with confidence-based filtering (six or more model consensus), preliminary backtesting yields a Sharpe ratio of 1.2 compared with 0.8 for buy-and-hold. Nonetheless, realistic deployment must account for transaction costs, slippage, and market impact, all of which can materially reduce performance even at low per-trade costs.

Several limitations warrant further investigation. Our evaluation focuses on U.S. equities between 2020–2023, leaving generalization to other market regimes and asset classes uncertain. The quantum component relies on classical simulation rather than physical quantum hardware, and all features are price-derived, omitting alternative data now central in quantitative finance. Future work should extend validation across broader datasets, incorporate additional data modalities (e.g., macroeconomic indicators or social sentiment), explore deployment on emerging quantum processors, and develop interpretability tools suitable for regulatory environments.

Overall, the results show that incremental but meaningful gains remain achievable in financial prediction through systematic integration of complementary techniques. The key insight is not a single algorithmic advance, but the coordinated combination of quantum features, transformer-based sequence modeling, and disciplined model filtering. For practitioners, architecture diversity and strict exclusion of weak predictors are critical for building high-performance ensembles. For researchers, this work highlights promising directions at the intersection of quantum-classical hybrid systems, transformer-based forecasting, and the theoretical foundations of financial ensemble learning, offering a rich agenda for continued exploration.

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